

Tuba Masood

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PROFESSIONAL SUMMARY

MS in Data Science candidate at Fitchburg State University seeking a paid Summer 2026 internship in data science or analytics. Holds a B.E. in Computer Science with hands-on experience in Python, Machine Learning, SQL, and full-stack development. Demonstrated ability to analyze complex problems, build data-driven solutions, and communicate technical insights clearly. Brings a unique combination of academic training, real-world software development, and 4+ years of teaching experience in programming and algorithms.

TECHNICAL SKILLS

Languages: Python, SQL, Java, C/C++, JavaScript

Data Science & ML: Machine Learning, Statistical Analysis, Data Wrangling, EDA, Pandas, NumPy, Scikit-learn

Data Visualization: Matplotlib, Seaborn, Tableau (familiar)

Databases: MySQL, Database Design, Query Optimization

Tools & Platforms: Jupyter Notebook, Git, GitHub, Google Colab, VS Code

Frameworks: ReactJS, Spring Boot, Bootstrap

Soft Skills: Analytical Thinking, Communication, Team Leadership, Teaching & Mentoring

EDUCATION

M.S. in Data Science | Fitchburg State University 2025 – Present | Fitchburg, MA

Relevant Coursework: Machine Learning, Data Mining, Statistical Methods, Big Data Analytics, Data Visualization, Database Systems

B.E. / B.Tech in Computer Science | Osmania University 2017 – 2021 | Hyderabad, India

Relevant Coursework: Advanced Machine Learning, Data Structures & Algorithms, Database Management Systems, Python Programming, OOP (Java), Cloud Computing, Software Engineering

PROJECTS

Healthcare Analytics: Chronic Disease Management Prioritization | Fitchburg State University — MS Data Science, 2025

- Designed and executed an end-to-end data science pipeline to determine optimal implementation sequencing for 5 chronic disease programs (Diabetes, Hypertension, CHF, COPD, Asthma) representing roughly \$569B in combined annual U.S. healthcare costs.
- Compiled and engineered a six-variable dataset (five used in the final composite score) from CDC, CMS, MEPS, and HCUP; applied Z-score normalization to eliminate scale bias across dollar and percentage variables.
- Performed EDA including correlation heatmaps — identified Annual Cost ($r = -0.87$) and QALE Loss ($r = -0.79$) as the strongest predictors; found each top-ranked condition was driven by a different dominant variable (cost, prevalence, or mortality).
- Built a Multi-Criteria Decision Analysis (MCDA) model that reproduced the established clinical priority ranking (Spearman's $Rho = 1.00$, MAE = 0.00 on $n=5$); cross-checked with KNN ($k=1$) and Decision Tree under LOOCV, achieving Spearman's Rho of 0.72 and 0.67 respectively.
- Conducted sensitivity analysis across 20 MCDA weight perturbations ($\pm 10\%$), with the full ranking stable in 13/20 configurations and the bottom two positions stable across all 20.

Tools: Python, Scikit-learn, Pandas, NumPy, SciPy, Matplotlib, Seaborn, Jupyter Notebook | **Methods:** MCDA, KNN, Decision Tree, Spearman/Kendall Correlation, LOOCV

PROFESSIONAL EXPERIENCE

Computer Science Instructor | St. Mathews Degree College Jun 2023 – Oct 2024 | Hyderabad, India

- Taught C/C++, Java, and Database-SQL covering theory and practical labs for 100+ undergraduate students.

- Designed curriculum with hands-on SQL exercises, reinforcing relational database and query optimization concepts directly applicable to data roles.
- Mentored students on algorithmic thinking and data structures, laying analytical foundations essential for data science work.

Software Developer | BeCoder

Apr 2021 – Dec 2022 | Hyderabad, India

- Built data-driven front-end interfaces consuming Google APIs, transforming API responses into actionable visual dashboards for clients.
- Developed interactive web applications using JavaScript and ReactJS, improving data presentation and user experience.

Software Developer (Flutter) | GAO-Tek Inc.

Feb 2023 – May 2023 | Remote

- Developed Android application UI integrating Google APIs and navigation systems, handling real-time data fetching and rendering.

WordPress Developer | GAO-Tek Inc.

Mar 2023 | Remote

- Built and deployed e-commerce websites for textiles, portfolios, and education sectors with structured data management.

Computer Tutor | Saif Sir's The Career Academy

Jan 2019 – Aug 2023 | Hyderabad, India

- Delivered in-depth instruction in C/C++, Java, Python, and Data Structures & Algorithms to ICSE and degree-level students.
- Developed strong communication and mentoring skills through years of breaking down complex technical concepts for diverse learners.

CERTIFICATIONS & COURSES

- Cisco DevNet Associate – Cisco (2021)
- Java Full Stack Development – Udemy (2023)
- Web Development Full-Stack – DevTown (2024)
- Web Development Course – Udemy (2021)
- PHP and MySQL – Udemy (2020)

ACHIEVEMENTS

- Participated in HACKATHONv1.0, Hyderabad (2020)
- Participated in Game of Codes (GOC) competition (2019)